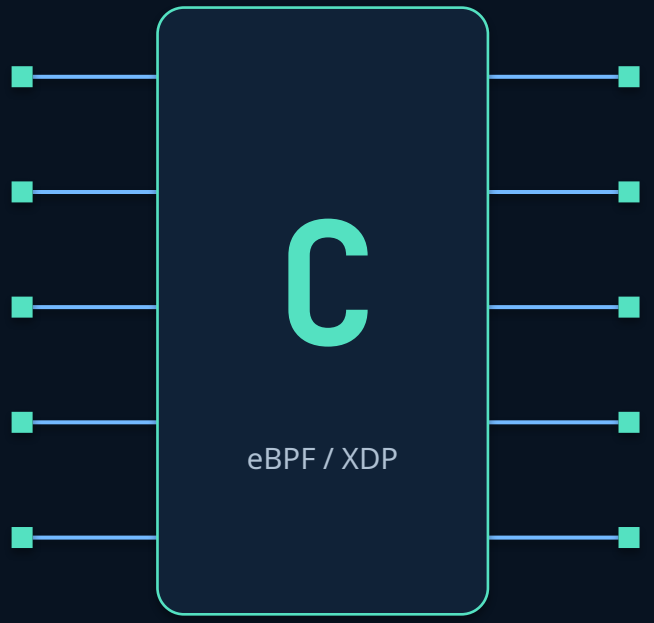
Project CIPHER

Behavioral protection for small networks

Designed to complement an existing firewall

THE 4 HORSEMEN



Allowed traffic can still be harmful.

LEGITIMATE ACTIVITY

- A camera uploads a backup

COMPROMISED DEVICE

- A camera uploads stolen data

Both may use allowed HTTPS. Context matters.

Add behavioral signals to existing protection.

PROPOSED DEPLOYMENT

Existing firewall

- Baseline access policy

CIPHER

- Device-level anomalies

Operator

- Monitor, review + trust

Blocking is established. Detection must add value.

CAPABILITY	EXISTING TOOLS	CIPHER TODAY
Packet rules	Standard firewall function	Implemented
Device anomalies	Requires added analysis	Experimental
XDP acceleration	Used by existing IPS	Generic mode
Comparative advantage	Established alternatives	Unproven

Sources: nftables and Suricata documentation; local code review.

Keep packet handling separate from analysis.

DATA

eBPF / XDP

- Kernel enforcement

›

EDGE

Rust daemon

- Metadata aggregation

›

SERVER

Axum server

- Statistical analysis

Generic XDP today. Throughput and latency require benchmarks.

Test whether AI improves decisions.

Z-score screening

- Four telemetry features
- Saved baseline + reset

LLM review

- Baseline + recent reports
- Advisory by default

Anthropic + compatible APIs. Offline booth uses scripted responses.

Alerts stay visible. Operators stay in control.

REHEARSED SOFTWARE FLOW

- See unusual activity and its evidence
- Enable response, then trust the device
- Show provider failure and recovery

DEMO

simulated sensor

`real backend`

Start with small offices we can support.

PILOT PROFILE

- Mixed office + IoT devices
- Compatible Linux gateway

TEST THE NEED

- Actionable alerts
- Manageable support burden

Customer demand and willingness to pay remain unvalidated.

Prove added protection at an acceptable cost.

Security

- Missed threats
- False blocks

Efficiency

- CPU / RAM
- Throughput / p99 latency

Operations

- Installation effort
- Failure recovery

Same hardware: firewall alone → + statistics → + LLM review

Ready to demonstrate. Hardware proof is next.

SHOW TODAY

- Alerts, trust + incident review
- Acknowledged responses + recovery

VALIDATE NEXT

- Live provider account + model
- Real XDP packet blocking

Synthetic traffic • real backend • scripted offline supervisor

The 4 Horsemen

01
Networks
- Deployment + testing

02
eBPF
- Packet enforcement

03
Rust
- Edge + server

04
AI
- Behavioral analysis

Help us earn the next deployment.

01

Design partners

- Define SOHO pilot needs

02

Hardware feedback

- Review edge/server setup

03

Mentors

- Validate security + reliability

Rehearsed demo → gateway validation → measured pilots